

Question	Answer	Marks	Guidance
2(a)	gas which obeys $pV \propto T$ (under all conditions)	M1	Allow $pV = nRT$ or $pV = NkT$ for $pV \propto T$.
	where T is the thermodynamic temperature	A1	Allow 'absolute' or 'Kelvin' for 'thermodynamic'.
2(b)(i)	measures absolute temperature (unlike liquid in glass) or much greater range of temperature (than liquid in glass) or gives a measurement that does not depend on the properties of a (real) substance	B1	Allow reverse argument. Allow 'thermodynamic temperature' Ignore 'Kelvin' Allow 'liquid in glass only accurate at fixed points'. Allow 'measures higher / lower temperatures'.
2(b)(ii)	very small mass of substance or rapidly changing temperatures	B1	Allow 'amount' or 'volume' for 'mass'. Allow remote reading using a thermocouple. Ignore 'small changes in temperature'
2(c)	$357^2 + 424^2 + 509^2 + 286^2 + 435^2 (= 837\,327)$	C1	Value 167 465 or 915 implies C1.
	$c_0 = \sqrt{\left(\frac{837\,327}{5}\right)}$ $= 409 \text{ m s}^{-1}$	A1	Must be given to 3 significant figures. AFC. Answer given to more than 3SF, that rounds to 409, implies the C1 mark.
2(d)(i)	$N = n \times N_A$	C1	
	$n = (5.0 \times 10^{23}) / (6.02 \times 10^{23})$ $= 0.83 \text{ mol}$	A1	Correct to at least 2 significant figures. AFC. (0.831)

Question	Answer	Marks	Guidance
2(d)(ii)	$\frac{1}{2}m \langle c^2 \rangle = (3/2) kT$	C1	Allow $pV = NkT$ <u>and</u> $p = \left(\frac{1}{3}\right)\left(\frac{Nm}{V}\right) \langle c^2 \rangle$. Allow $pV = nRT$ <u>and</u> $p = \left(\frac{1}{3}\right)\left(\frac{Nm}{V}\right) \langle c^2 \rangle$ Allow $\langle c^2 \rangle = \left(\frac{3kT}{m}\right)$ Allow c_{rms} for $\sqrt{\langle c^2 \rangle}$ Ignore $\langle c \rangle^2$
	$T_0 = [(3.34 \times 10^{-27}) \times 409^2] / (3 \times 1.38 \times 10^{-23})$ $= 13.5 \text{ K}$	A1	Possible ECF from 2(c) or 2(d)(i). Correct to at least 3 significant figures. AFC. (13.51)
2(d)(iii)	line <u>starting</u> at (T_0, c_0) with positive gradient at that point	B1	
	line with decreasing positive gradient	B1	Allow line becoming horizontal at the end but not negative gradient.
	line that passes through $(0, 0)$ when extrapolated back from $T = T_0$	B1	Ignore any line drawn by candidate before T_0. Extrapolate candidate's line from T_0 and beyond back to $T = 0$. Award MP3 if there is a conceivable reasonable extension of their line that passes through the test box centred on the origin, without having to resort to incorrect curvature.